

Class is made of two things-

(i) characteristics.

(ii) functionalities.

Characteristics are - Member variables of the class

Functionalities are methods of the class.

These two components can be public, private, protected

Symbol for - Access modifiers

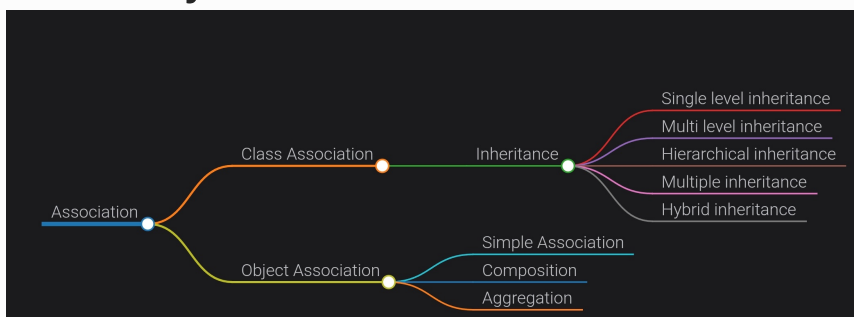
private	→	-	("minus")
protected	→	#	("Hash")
public	→	+	("plus")

Association

Association provides a mechanism to communicate one object with another object, or one object provides services to another object. Association represents the relationship between classes.

The association can be divided into two categories:

- Class association (Inheritance)
- Object association



Class level Association is Inheritance

Two class can be associated only as a "is-a" relation.

Representation of "is-a" relation in class diagram is
"→" [closed arrow]

Object level association is 3 type:-

1. Simple Association

2. Aggregation

3. Composition

Simple association:- It is a very weak relation between two objects.

Example:- Ayon lives in a house.

Associated

but they are not strongly associated since Ayon does not own the house.

Simple association is called "has-a" relation.

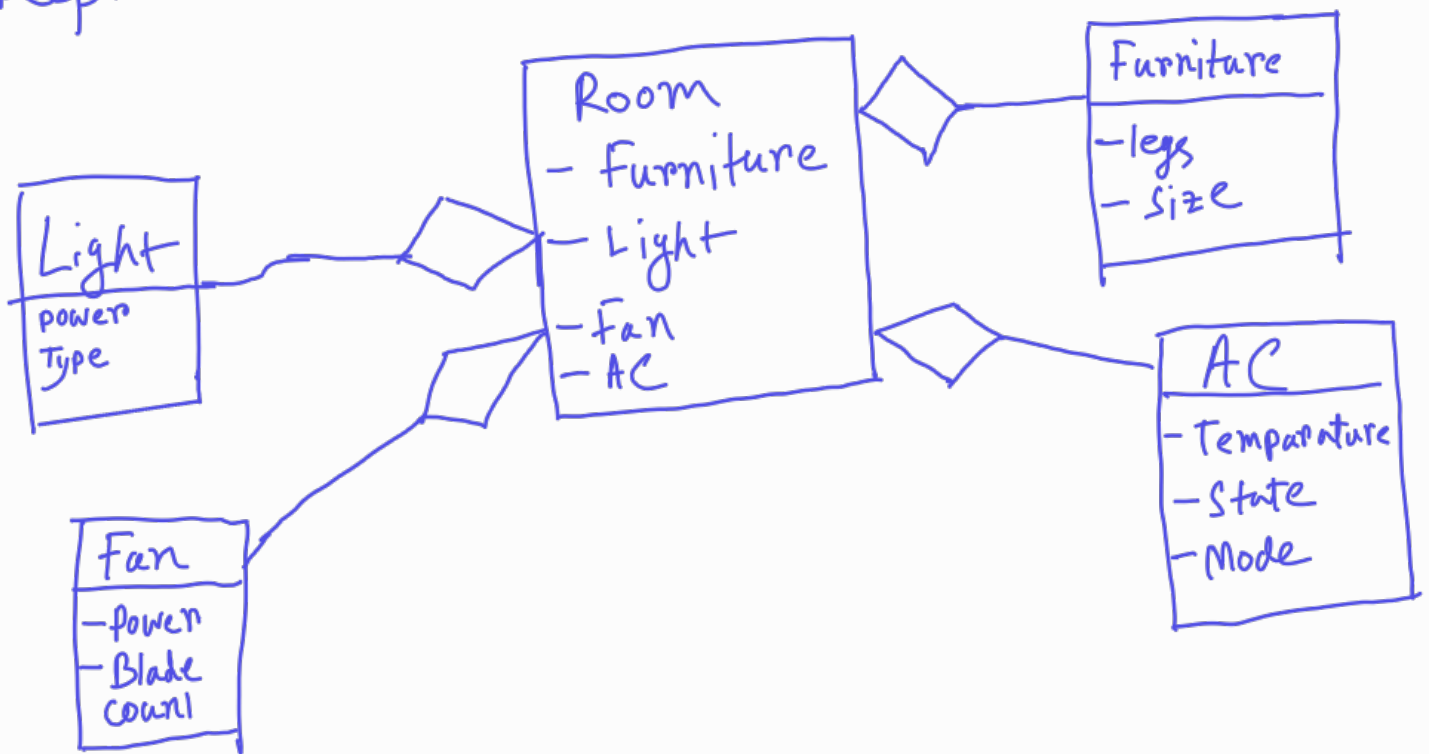
Representation in class diagram of "has-a" is "→" [open arrow]

Aggregation:- Aggregation shows the relation between A container object and the objects it contains.

This association is also not a strong association.

Example:- A living ^{room} contains (has) sofa, fan, light, A.C.
but they can move and replace to other room.

Representation in class diagram — "◊" [Unfilled Barfi]



Composition:- Composition is a relation between An object and the object that it is made of.

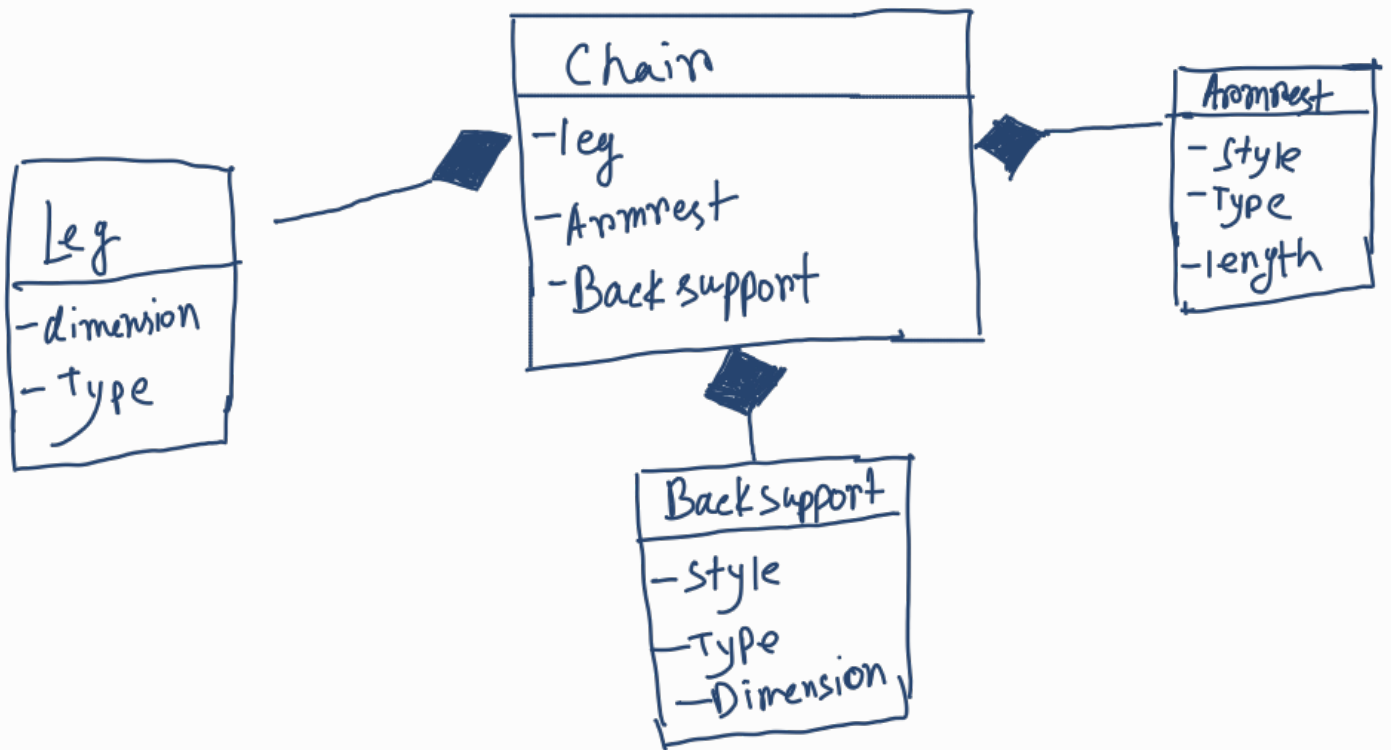
It is the relation between part objects & the whole object

Example:- A normal chair is made of legs, seat, back rest, handrest.

legs, seat, back rest, handrest these cannot exist without chair.

So, chair (whole object) is strongly related with those parts.

Diagram representation:- "◆" [filled barfi]



Handbook of presentations in class-Diagram

is-a (class level association): "→" [closed arrow]

has-a (Object level weak association): "→" [open arrow]

Aggregation (Association of Container object & object it contains):
"◻" [unfilled barfi]

Composition (Association of whole object & parts of which it is made of):
"◆" [filled barfi]

Access modifier -

public: + "plus"

private: - "minus"

protected: # "hash"
